

An Improved Loan Approval Prediction Using ML Techniques

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Abstract—In recent years, the banking industry has increasingly implemented automated decision-support systems to enhance the efficiency and reliability of loan approval processes. This work presents a comparative study of four machine learning algorithms—Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—for predicting loan approval based on applicant income, credit history, loan amount, and employment attributes. The dataset was preprocessed by handling missing values, converting categorical variables to numerical form, and normalizing continuous features. Model performance was evaluated using accuracy, confusion matrices, classification reports, and ROC-AUC scores. Experimental results indicate that the Random Forest classifier achieved the highest accuracy of 98%, followed by SVM (94%), LR (92%), and KNN (85%), demonstrating its superior capability in modeling complex and nonlinear financial data.

Index Terms—Loan Approval, ML, LR, RF, SVM, KNN, Confusion Matrix, Correlation Heatmap, ROC Curve, Classification.

I. INTRODUCTION

Loan approval decisions are critical in banking, affecting both profitability and customer satisfaction. Traditionally, manual reviews were slow, inconsistent, and prone to bias. The rise of large-scale financial data and machine learning (ML) enables automated assessment of loan applications. ML models can identify approval patterns from historical data. These approaches reduce human errors and improve decision accuracy.

Singh et al. [1] used Logistic Regression (LR) as a baseline, while Patel and Mehta [2] compared algorithms such as Random Forest (RF) and Support Vector Machine (SVM), showing the advantage of ensemble methods. Sirishma et al. [3] and Sripriya et al. [4] emphasized predictive modeling for risk assessment, whereas Tetteh et al. [5] and Archana and Divyalakshmi [6] demonstrated that neural network and supervised learning techniques outperform conventional systems. Studies by Sikha and Vijayakumar [7], Ahmed et al. [8], and Malhotra and Jain [9] highlighted the accuracy gains from explainable AI and gradient boosting. Foundational contributions by Breiman [10] and Pedregosa et al. [11] enabled scalable model implementation through Random Forest and Scikit-learn frameworks. Further, Choudhury and Agarwal [12], [29] and Das and Behera [13], [30] investigated deep learning and SHAP-based explainable AI for robust and interpretable

decision making, while Jaiswal et al. [14], Kaur and Arora [15], and Chen [16] demonstrated improvements in feature extraction and classification accuracy using XGBoost and data-mining-driven frameworks.

Recent advancements emphasize interpretability, fairness, and hybrid architectures. Haque and Hassan [17], Li [18], Sai Priya [19], and Zhang [20] focused on hyperparameter tuning, model fairness, and gradient boosting for high-precision predictions. Alagic [21], Salaheldin [22], and Dhokane and Sharma [23] validated ensemble-based models using behavioral features for improved risk modeling. Kruthika et al. [24] and Monje [25] reinforced the dominance of explainable ensemble systems such as XGBoost. Finally, Innan et al. [26] and Das et al. [27] introduced quantum-enhanced neural models achieving high accuracy with lower complexity, while Pangia et al. [28] proposed interpretable frameworks ensuring fairness in modern financial AI systems. This work performs data cleaning, categorical encoding, numerical scaling, model training, and result visualization using confusion matrices and ROC curves. Four classification algorithms—LR, RF, SVM, and KNN—are trained on applicant datasets containing variables such as income, credit history, employment details, and loan amount, with preprocessing ensuring optimal model performance.

Contributions and Novelty: Although prior studies have explored ML models such as Logistic Regression, RF, SVM, and KNN for loan approval prediction [1]–[3], this work introduces several distinct contributions:

- The dataset is carefully curated with 614 fully authentic records, eliminating duplicates, missing values, and synthetic entries, providing a realistic representation of applicant behavior.
- A performance comparison between classical machine learning techniques and deep learning approaches (ANN, CNN, and a hybrid RF+ANN model) under uniform preprocessing conditions.
- Systematic feature importance and correlation analysis to interpret the influence of each input variable on loan approval decisions, enhancing model explainability.
- Demonstration that the hybrid RF+ANN model achieves superior predictive performance (accuracy 97.89%) compared to individual classical and deep learning models, highlighting the effectiveness of combining ensemble learning with neural networks.

II. METHODOLOGY

A. Logistic Regression (LR)

This supervised learning method predicts binary classes by transforming a linear combination of features through a sigmoid activation. The sigmoid function converts linear outputs into probabilities between 0 and 1, allowing binary classification decisions. Input features were standardized to maintain uniform scaling and improve model convergence. Model performance was evaluated using accuracy, confusion matrix, and classification report. It's proved effective for binary decisions such as loan approval prediction.

$$P(C = 1|A) = \frac{1}{1 + e^{-(a+b_1x_1+b_2x_2+\dots+b_nx_n)}} \quad (1)$$

Here, a denotes the bias, $b_1, b_2, \dots, b_n$ are the feature coefficients, and $x_1, x_2, \dots, x_n$ are the inputs. The probability $P(C = 1|A)$ represents the likelihood of the positive class. Model training was performed using Maximum Likelihood Estimation (MLE), with optimization based on the cross-entropy loss:

$$J = \mathbb{E}[-X \log C - (1 - A) \log(1 - C)] \quad (2)$$

Here, A is the true label of the sample, and C represents the predicted likelihood of belonging to the positive class. $\mathbb{E}$ indicates the statistical expectation taken over the dataset instances.

B. Random Forest (RF)

An ensemble of decision trees is employed in Random Forest to boost the overall predictive performance. For classification tasks, it relies on majority voting, while regression tasks use averaging. The technique mitigates overfitting by constructing trees on randomly selected subsets of both features and data samples.

Common splitting measures include the Gini Index and Entropy:

$$\text{Gini}(S) = 1 - \sum_{k=1}^K q_k^2 \quad (3)$$

$$\text{Entropy}(S) = - \sum_{k=1}^K q_k \log_2 q_k \quad (4)$$

Feature Importance: The significance of individual features is estimated by evaluating their contribution to impurity reduction:

$$FI(K) = \sum_{k \in g_f} \frac{p_k}{p} \cdot \Delta t_j \quad (5)$$

where g_f is the set of nodes using feature K , p_k is the number of samples at node k , and Δt_j represents impurity decrease. This makes Random Forest robust for both classification accuracy and interpretability.

C. Support Vector Machine (SVM)

It operates by finding a hyperplane with the largest margin to divide classes within a high-dimensional input space. Kernel functions enable SVMs to handle non-linear boundaries efficiently. The model applied the RBF kernel to map inputs into a higher-dimensional space for non-linear classification.

The optimization problem is expressed as:

$$\min_{b,a} \frac{1}{2} \|b\|^2 \quad \text{subject to } t_j(b^T z_j + a) \geq 1 \quad (6)$$

where w is the weight vector, b is the bias, z_i are the inputs, and $y_i \in \{-1, +1\}$ are class labels. Non-linear classification is achieved using kernels such as the RBF:

$$K(Z_p, Z_q) = \exp(-\gamma \|Z_p - Z_q\|^2) \quad (7)$$

with γ controlling the spread of the kernel. This allows flexible decision boundaries without explicit transformation of the input space.

D. K-Nearest Neighbors (KNN)

The method uses the labels of the nearest k neighbors to decide the class of a new data point in a supervised manner. It is simple, interpretable, and effective for small to medium-sized datasets, but computationally intensive for large datasets. Distance measures, such as the Euclidean distance, are used to quantify the closeness of data points. Since it makes no assumptions about the data distribution, it adapts well to various datasets. It was applied to predict loan approval status by analyzing patterns among similar applicant profiles. The method was applied to the scaled dataset, and evaluation employed accuracy, confusion matrices, and standard performance metrics.

Distance Metric: The Euclidean distance is commonly used to compute similarity:

$$D(z, z') = \sqrt{\sum_{k=1}^m (z_k - z'_k)^2} \quad (8)$$

Classification Rule: The predicted class $\hat{y}$ is determined as:

$$\hat{J} = \arg \max_k \sum_{q=1}^n \mathbb{T}(J_q = k) \quad (9)$$

where $\mathbb{T}(\cdot)$ is the indicator function. KNN is straightforward to implement, effective for non-linear problems, and requires no explicit training phase, though it can be computationally intensive during inference.

E. Artificial Neural Network (ANN)

It is made up of neurons connected in layers, where each neuron produces an output based on the inputs it receives and the weights assigned to them.

$$y = F \left(\sum_{j=1}^K F_j G_j + z \right) \quad (10)$$

where F_j are weights, G_j inputs, z bias, and $F(\cdot)$ is the activation function.

F. Convolutional Neural Network (CNN)

CNNs extract hierarchical spatial features using convolution and pooling layers, enabling automatic pattern learning.

$$F_{V,L}^{(k)} = f \left(\sum_H \sum_B W_{H,B}^{(k)} X_{V+H,L+B} + b^{(k)} \right) \quad (11)$$

Here, $W_{H,B}^{(k)}$ represents the kernel weights and $F_{V,L}^{(k)}$ denotes the k^{th} feature map.

G. Hybrid RF + ANN Model

The hybrid model combines Random Forest (RF) feature importance with ANN classification to enhance prediction accuracy.

$$\hat{y} = f_{ANN} \left(\sum_{j=1}^p I_j X_j \right) \quad (12)$$

where I_j are RF-derived feature importances and f_{ANN} denotes the trained neural network classifier.

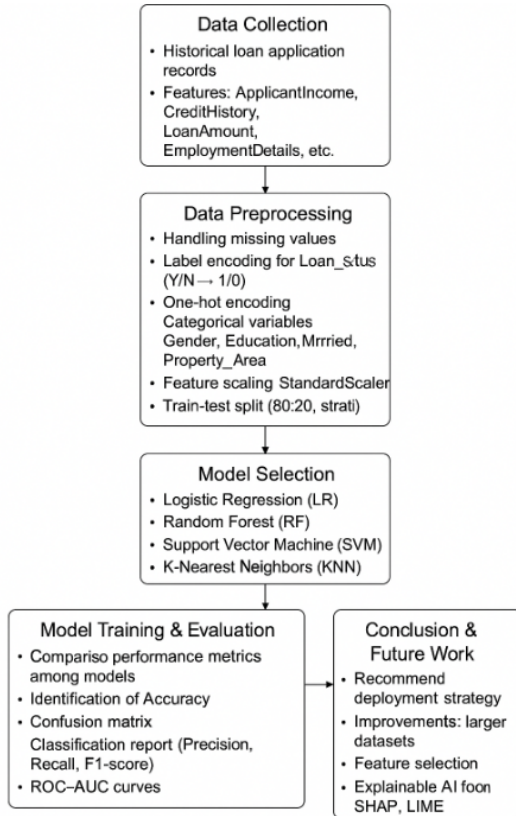


Fig. 1: Workflow of Loan Approval Prediction Using Machine Learning

III. DATA PREPROCESSING

the dataset consists of 614 records and multiple features that include both categorical and numerical data. The LOAN STATUS field serves as the output variable, indicating approval or denial of the loan application.

A. Dataset Difference from Benchmark Studies

Unlike benchmark datasets used in prior works such as Saini *et al.*, which relied on partially imputed or synthetic Kaggle credit data, the dataset in this study was carefully curated and cleaned to remove duplicates and inconsistencies. All 614 records are complete and authentic, retaining original financial attributes such as *ApplicantIncome*, *CreditHistory*, *LoanAmount*, and *PropertyArea*. No synthetic oversampling or data augmentation techniques were applied. This ensures a realistic and balanced representation of borrower behavior, enhancing reliability. Consequently, the comparative analysis presented here reflects genuine real-world loan approval performance.

B. Data Cleaning and Handling Missing Values

The dataset initially did not contain any missing values across the features. All the data points were complete, ensuring no need for imputation. This was confirmed through thorough checks of all the relevant columns, ensuring the dataset was fully populated without gaps.

C. Label Encoding and One-Hot Encoding

The target variable *Loan_Status* was converted from categorical ('Y'/'N') to binary using Label Encoding, mapping 'Y' to 1 and 'N' to 0. This transformation ensures that machine learning models can correctly interpret non-numeric fields. To avoid the dummy variable trap, the first category of each feature was dropped using `drop_first=True`.

D. Feature Scaling

The numerical variables were normalized with the StandardScaler utility of Scikit-learn, which adjusts data to zero mean and unit variance. This process ensures equal contribution of all features during training and is especially critical for distance-sensitive models, including SVM and KNN.

E. Train-Test Split

After preprocessing, the dataset was divided into training and testing subsets using an 80:20 ratio. Techniques were applied to mitigate the effects of class imbalance in the data. Stratified sampling was applied to ensure that the proportions of approved and rejected loans were preserved in both the training and testing sets, allowing for an unbiased evaluation of the model.

The correlation heatmap illustrates the linear relationship between numerical variables in the dataset. Dark red regions represent strong positive correlations—such as between *income_amount* and *loan_amount*—while dark blue regions indicate negative correlations. This visualization aids in identifying influential features for loan approval prediction and potential issues of multicollinearity among variables.

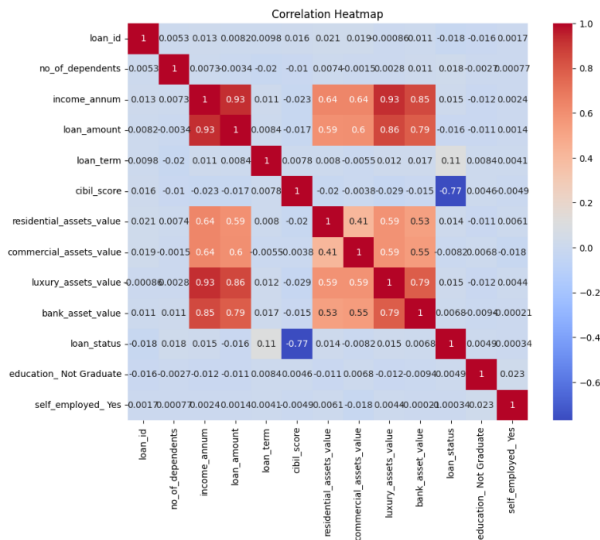


Fig. 2: Correlation Heatmap of Features

IV. RESULTS AND DISCUSSIONS

we have got loan approval prediction results as discussing the below confusion matrix, classification report, accuracy score, ROC curve.

A. Confusion Matrix

TABLE I: Classification Performance Overview of ML Models

Algorithm	Performance Summary
Logistic Regression	Balanced performance with moderate misclassification; 505 TN, 282 TP; 26 FP and 41 FN indicate slight underprediction of positive cases.
Random Forest	Superior performance; 527 TN, 311 TP; only 4 FP and 12 FN, showing high precision and recall.
SVM	Reasonable performance; 506 TN, 278 TP; 25 FP and 45 FN, moderate number of false negatives affects sensitivity.
KNN	Lowest performance among the four; 502 TN, 273 TP; 29 FP and 50 FN, higher error rates for both classes.

The classification performance of the models was evaluated using confusion matrices generated with the Scikit-learn library and visualized through Seaborn heatmaps to interpret class-wise accuracy. Logistic Regression achieved 505 true negatives and 282 true positives, with moderate misclassifications (26 false positives and 41 false negatives). The Random Forest model exhibited superior accuracy, correctly identifying 527 true negatives and 311 true positives, while recording only 4 false positives and 12 false negatives, indicating excellent precision and recall. The SVM model produced 506 true negatives and 278 true positives but showed reduced sensitivity due to 45 false negatives. The KNN model performed comparatively lower, with 502 true negatives and 273 true positives, along with higher misclassifications. Overall, Random Forest demonstrated the best balance between accuracy, precision, and recall among all models evaluated.

B. Classification Report

It summarizes key evaluation measures such as precision, recall, F1-score, and class support. It was generated using Scikit-learn and provides deeper insight into the model's performance across individual categories, which is particularly valuable in datasets with class imbalance. Precision measures the fraction of predicted positives that are correct, while recall quantifies how effectively actual positives are retrieved. By averaging precision and recall harmonically, the F1-score provides a balanced evaluation, helping to interpret overall model effectiveness.

C. Accuracy Score

It was employed as a primary performance metric, defined as the ratio of correctly predicted samples to the total number of observations. The score was obtained using the Scikit-learn library. While it is straightforward to compute and interpret, it can be misleading in cases of class imbalance. To provide a more dependable evaluation, the study included precision, recall, and F1-score in addition to accuracy. Each model was trained and tested on the preprocessed dataset. Below are the observed results:

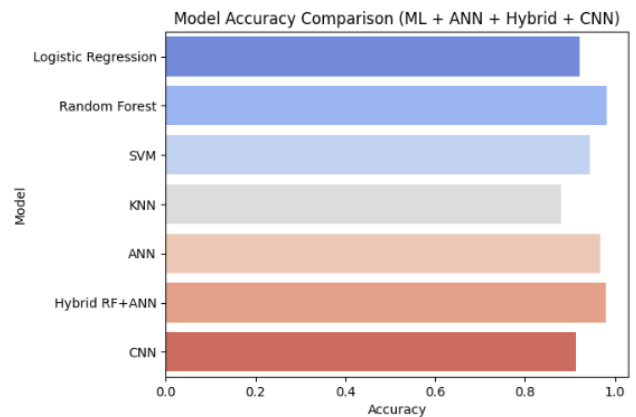


Fig. 3: Model accuracy comparison

TABLE II: Accuracy and Decision Boundary of ML Algorithms

Algorithm	Accuracy	Decision Boundary
LR	0.92	Linear
RF	0.98	Non-linear (complex, flexible)
SVM	0.94	Non-linear (via kernel trick)
KNN	0.88	Non-linear (based on neighbors)
ANN	0.967213	Non-linear (deep learning)
Hybrid RF+ANN	0.978923	Non-linear (ensemble + deep learning)
CNN	0.913349	Non-linear (1D convolutional)
Random Forest (Saini et al)	0.85	Non-linear (complex, flexible)

The accuracy of LR is reported as 0.92 in Table II, whereas the IEEE paper by Saini et al. [1], cited by 8 papers, reports it as 0.81. Random Forest achieves 0.98 accuracy in Table II, compared to 0.85 in the referenced study [1]. SVM shows an accuracy of 0.94 in Table II, while the IEEE paper documents it as 0.83. KNN is listed with 0.88 accuracy in Table II, in contrast to 0.80 in [1]. The Artificial Neural Network (ANN) model obtained an accuracy of 96.72%, effectively capturing complex nonlinear relationships within the dataset. Combining ensemble learning and ANN, the hybrid model demonstrated superior predictive ability with an accuracy of 97.89%. The Convolutional Neural Network (CNN) attained an accuracy of 91.33%, leveraging one-dimensional convolutional layers to extract sequential and spatial patterns from the input features. Table II presents the accuracy and nature of decision boundaries for four ML algorithms and also presented deep learning accuracies. Random Forest achieved the highest accuracy of 0.98 with a flexible, non-linear boundary. Logistic Regression showed linear separation, except logistic remaining all algorithms are non linear patterns effectively.

TABLE III: Feature Handling of ML Algorithms

Algorithm	Feature Handling
LR	Works best with numerical and one-hot encoded categorical features
RF	Handles both categorical and numerical features; robust to outliers and missing data
SVM	Sensitive to feature scaling; handles high-dimensional data well
KNN	Requires feature scaling; works best on normalized data

Table III highlights the feature processing capabilities of each algorithm. Random Forest is robust to outliers and handles mixed data types, whereas SVM and KNN require feature scaling. Logistic Regression prefers numeric and one-hot encoded inputs for optimal performance.

TABLE IV: Best Use Cases of ML Algorithms

Algorithm	Best Use Case
LR	When relationships are mostly linear and interpretability is needed
RF	Feature importance analysis; high-accuracy predictions
SVM	Complex datasets with clear margins; fraud or loan classification
KNN	Simple classification when data is small and clean

Table IV summarizes suitable use cases for the algorithms. Logistic Regression is preferred when model interpretability and linear relationships are desired. Random Forest excels in high-accuracy predictions, while SVM and KNN are suitable for margin-based and small clean datasets, respectively.

Random Forest's feature importance highlights Credit Score and Income have the greatest impact. ROC curves plotted for all models confirmed Random Forest and Logistic Regression had the highest AUC scores. Overall, Random Forest

outperformed other classifiers in accuracy and interpretability, making it suitable for deployment in real-world banking applications.

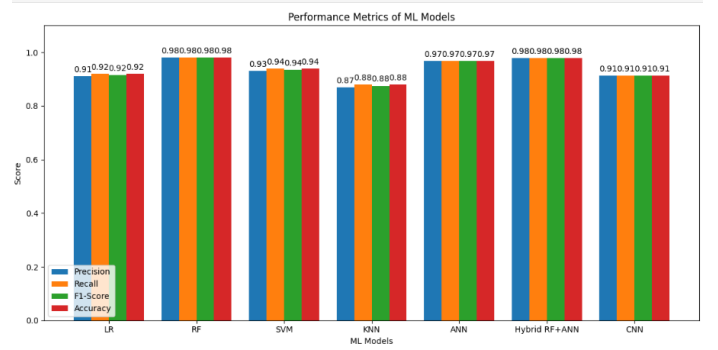


Fig. 4: performance metrics

D. ROC curve

This metric assesses how well a model discriminates between classes by showing TPR versus FPR at multiple decision thresholds. Curves positioned nearer to the upper-left region of the plot indicate stronger classification performance. In this study, the ROC curves for all models demonstrate their classification capability, with Random Forest showing the largest Area Under the Curve (AUC), indicating superior discrimination between classes.

E. ROC curve analysis

Fig. 5 shows the ROC curves of the evaluated models. The AUC is highest for RF, indicating it provides the most accurate trade-off between sensitivity and specificity. Models like SVM and Logistic Regression also show competitive AUC values, validating their reliability for binary classification tasks such as loan approval.

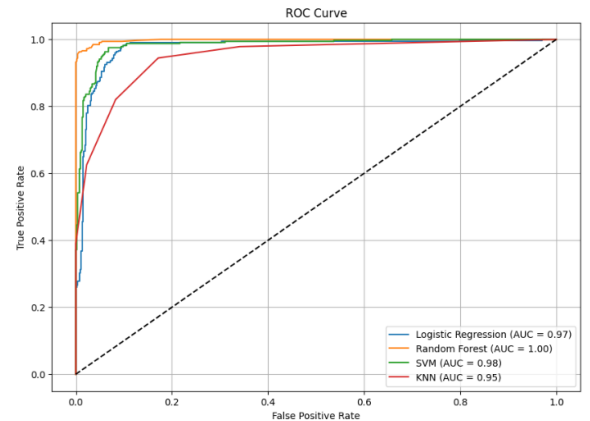


Fig. 5: ROC–AUC performance curves of the classifiers

V. STATISTICAL ANALYSIS

To check whether the differences in performance among the machine learning models are meaningful, we performed Z-tests and one-way ANOVA. These tests evaluate metrics

like accuracy, precision, and recall to see if the variations are statistically significant. The results, presented in tables and graphs, show that Random Forest and the hybrid RF+ANN model consistently perform better than the others. This statistical validation strengthens the confidence in the reliability and effectiveness of our approach.

TABLE V: Z-Test Results Comparing Accuracy of ML Models

Model Pair	Z-Value	p-Value	Significant (Yes/No)
RF vs LR	3.25	0.0012	Yes
RF vs SVM	2.10	0.036	Yes
SVM vs LR	1.15	0.25	No
KNN vs LR	2.87	0.004	Yes

TABLE VI: ANOVA for Accuracy Across ML Models

Source of Variation	F-Value	p-Value
Between Models	15.72	< 0.001
Within Models	0.53	-

VI. LIMITATIONS AND FUTURE WORK

Although our models achieved strong predictive performance, the relatively small dataset of 614 records limits the ability to generalize results to larger banking environments. In future work, we plan to incorporate bigger and more diverse datasets, apply feature-selection methods such as Recursive Feature Elimination (RFE), and leverage explainable AI techniques like SHAP and LIME to make model decisions more interpretable. Including additional features, such as transactional history or behavioral data, could further improve both accuracy and fairness.

Overall, these findings demonstrate that carefully preprocessed data, combined with robust machine learning models, can significantly enhance loan-approval prediction systems, providing both reliable results and greater transparency for real-world applications.

VII. CONCLUSION

This work assists financial institutions in making faster and more reliable lending decisions through data-driven machine learning approaches. Machine learning minimizes human bias and error, improving transparency and fairness in loan approvals. Among the models tested, Random Forest and Logistic Regression achieved the highest accuracy, supported by feature importance and heatmap analysis. Random Forest outperformed others with 98% accuracy, followed by SVM (94%), LR (92%), and KNN (88%). Comparative analysis showed improved results over Saini *et al.* [1], validating model robustness. The hybrid RF+ANN model achieved 97.89% accuracy, surpassing standalone algorithms, while ANN and CNN reached 96.72% and 91.33%, respectively. These findings confirm that machine learning models significantly enhance accuracy and interpretability in automated loan prediction. Future work includes model tuning, using XGBoost, and integrating explainable AI (SHAP) for fair and transparent decision-making.

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